

Student Learning Analytics System

1. Project Overview

Project Name: Student Learning Analytics System. | MySQL | PowerBI | SQL Analytics

Description:

Student Learning Analytics System designed to manage and analyze academic data using MySQL. The project demonstrates how a relational database can efficiently organize student records, course enrollments, attendance, assignments, examinations, and performance analytics.

The system is designed following relational database principles and normalized table structures to ensure data integrity and efficient querying.

2. Objective

The project simulates a real Learning Management System (LMS) used in educational institutions.

The system stores information related to

- Students
- Courses
- Instructors
- Attendance
- Assignments
- Examinations
- Results
- Performance Analysis

The collected data is analyzed using SQL queries to generate useful academic insights

3.Problem Statement

Educational institutions generate a large amount of student data every day.

However, most institutes struggle to answer questions like:

- Which students are at risk of failing?
- Which course has the highest enrollment?
- Which instructor has the best-performing students?
- Which students have poor attendance?
- What is the pass percentage of each course?
- Which students have not submitted assignments?

This system solves these problems using SQL.

4. Scope

The system will allow:

Student Management

- Register students
- Update student details
- Remove inactive students

Course Management

- Create new courses
- Assign instructors
- Update course information

Enrollment Management

- Enroll students
- Track enrollment status
- Monitor course completion

Attendance Management

- Mark attendance
- Calculate attendance percentage
- Identify irregular students

Assignment Management

- Create assignments
- Track submissions
- Evaluate assignment scores

Examination Management

- Schedule exams
- Store exam results
- Calculate averages

Performance Analytics

Generate reports for

- Top Performers
- Weak Students
- Course Performance
- Instructor Performance
- Monthly Trends

5. Functional Requirements

The system should support:

Student Module

- Add Student
- Edit Student
- Delete Student
- Search Student

Instructor Module

- Add Instructor
- Assign Instructor to Course

Course Module

- Add Course
- Update Course
- Delete Course

Enrollment Module

- Enroll Student
- Drop Student
- View Enrollment History

Attendance Module

- Mark Attendance
- Calculate Attendance %

Assignment Module

- Create Assignment
- Submit Assignment
- Evaluate Assignment

Examination Module

- Schedule Exam
- Store Marks
- Generate Result

Reporting Module

Generate reports such as

- Course-wise Performance

- Student-wise Performance
- Attendance Reports
- Pass Percentage
- Risk Analysis

6. Non-Functional Requirements

- Fast Query Execution
- Data Integrity
- Data Security
- Easy Maintenance
- Scalable Database Design

7. Technology Stack

Technology	Purpose
MySQL	Database
MySQL Workbench	Query Execution
Excel / CSV	Dataset
Git	Version Control
GitHub	Repository
Draw.io	ER Diagram
Power BI	Dashboard

8. Database Design : Tables

1. Students

Stores student information.

Column	Type	Description
student_id	INT	Primary Key
first_name	VARCHAR(50)	First Name
last_name	VARCHAR(50)	Last Name
gender	VARCHAR(10)	Gender
date_of_birth	DATE	DOB
email	VARCHAR(100)	Email

Column	Type	Description
phone	VARCHAR(20)	Phone
city	VARCHAR(50)	City
admission_date	DATE	Admission Date

2. Instructors

Stores faculty information.

3. Departments

Stores department information.

4. Courses

Stores all courses.

5. Enrollments

Many-to-many relationship between students and courses.

6. Attendance

Stores attendance records.

7. Assignments

Assignment details.

8. Assignment Submissions

Submission records.

9. Exams

Exam details.

10. Exam Results

Student marks.

11. Fees (*Optional but recommended*)

Tracks fee payment.

12. Student Risk Analysis (Unique Feature)

Stores calculated student performance indicators.

| student_id | attendance | average_marks | assignment_score | risk_level |

This table makes your project stand out because it simulates a real analytics feature used by institutions.

9. Database Relationships

Departments

|

Instructors

|

Courses

|

Enrollments

|

Students

|

Attendance

|

Assignments

|

Assignment Submissions

|

Exams

|

Exam Results

|

Risk Analysis

10. SQL Concepts Covered

Database

- CREATE DATABASE
- DROP DATABASE

Table Design

- CREATE TABLE
- ALTER TABLE
- DROP TABLE

Constraints

- Primary Key
- Foreign Key
- Unique
- Check
- Default
- Auto Increment

CRUD

- INSERT
- UPDATE
- DELETE

Retrieval

- SELECT
- WHERE
- DISTINCT
- LIMIT
- ORDER BY

Aggregation

- COUNT
- SUM
- AVG
- MIN
- MAX
- GROUP BY
- HAVING

Joins

- INNER JOIN
- LEFT JOIN

- RIGHT JOIN
- CROSS JOIN
- SELF JOIN

Advanced SQL

- CASE
- IF
- Subqueries
- Correlated Subqueries
- EXISTS
- ANY
- ALL
- UNION
- UNION ALL

Window Functions

- ROW_NUMBER()
- RANK()
- DENSE_RANK()
- NTILE()
- LAG()
- LEAD()

CTE

- WITH clause

Views

Create reusable reports.

Stored Procedures (Optional)

11. Business Questions

Your project should answer questions such as:

Student Analytics

- Who are the top 10 students?
- Who are the bottom 10 students?
- Which students have attendance below 75%?
- Which students have missing assignments?
- Which students are at high risk?

Course Analytics

- Which course has the highest enrollment?
- Which course has the highest average marks?
- What is the completion rate of each course?

Instructor Analytics

- Which instructor has the best-performing students?
- Which instructor teaches the most students?

Attendance Analytics

- Monthly attendance trend
- Students with continuous absences

Performance Analytics

- Monthly pass percentage
- Average marks by department
- Rank students within each course

12. Expected Outcomes

After completing this project, you will have:

- A fully normalized SQL database
- 10–12 related tables
- 500–1000+ sample records
- 60–100 SQL queries
- ER Diagram
- Data Dictionary
- Analytical Reports
- GitHub-ready documentation
- A strong portfolio project for Data Analyst interviews